



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Division: A

Subject: Programming with Java

ASSIGNMENT No : 16

TITLE : Write a Java program to create and handle a user-defined exception.

Problem Statement

Write a Java program to create and handle a user-defined exception.

Learning Objectives

To create and implement custom exceptions for application-specific error handling.

Theory

User-defined exceptions are created by extending the **Exception** class. They allow developers to define meaningful exceptions for specific business logic. Custom exceptions improve program readability and simplify debugging by providing descriptive error messages. They are widely used in enterprise software and validation systems.

Output Screenshots

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp16/Main.java
Something went wrong
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

CODE

```
1 package Exp16;
2
3 class MyException extends Exception {
4     MyException(String message) {
5         super(message);
6     }
7 }
8
9 public class Main {
10     public static void main(String[] args) {
11         try {
12             throw new MyException("Something went wrong");
13         }
14         catch (MyException e) {
15             System.out.println(e.getMessage());
16         }
17     }
18 }
```

EXERCISE 1

```
1 package Exp16.Exercise;
2
3 class InvalidAgeException extends Exception {
4     InvalidAgeException(String message) {
5         super(message);
6     }
7 }
8
9 public class Exp1 {
10     public static void main(String[] args) {
11         int age = 16;
12 }
```

```
13         try {
14             if (age < 18) {
15                 throw new InvalidAgeException("Age is below 18.
Not eligible to vote.");
16             }
17
18             System.out.println("Eligible to vote.");
19         }
20         catch (InvalidAgeException e) {
21             System.out.println(e.getMessage());
22         }
23     }
24 }
```

EXERCISE 2

```
1 package Exp16.Exercise;
2
3 class InvalidAgeException extends Exception {
4     InvalidAgeException(String message) {
5         super(message);
6     }
7 }
8
9 public class Exp2 {
10     public static void main(String[] args) {
11         int age = 20;
12
13         try {
14             if (age < 18) {
15                 throw new InvalidAgeException("Age is below 18.
Not eligible for driving license.");
16             }
17 }
```

```

18         System.out.println("Eligible for driving
license.");
19     }
20     catch (InvalidAgeException e) {
21         System.out.println(e.getMessage());
22     }
23 }
24 }

```

Exercise

Q1. Create a Voting System that throws a custom exception if age is below 18.

```

dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp16/Exercise/Exp1.java
Age is below 18. Not eligible to vote.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ █

```

Q2. Create a Driving License System that throws a custom exception if the user's age is below 18. If the age is valid, display that the user is eligible for a driving license.

```

dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp16/Exercise/Exp2.java
Eligible for driving license.
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ █

```

Conclusion

This experiment helped in understanding the concept of user-defined exceptions in Java. A custom exception can be created by extending the Exception class and can be used to handle application-specific errors. The experiment demonstrated how user-defined exceptions can make programs more readable, meaningful, and easier to debug, especially in validation-based applications.